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Physics 0525: Lab 7
Field-Effect Transistor (FET)

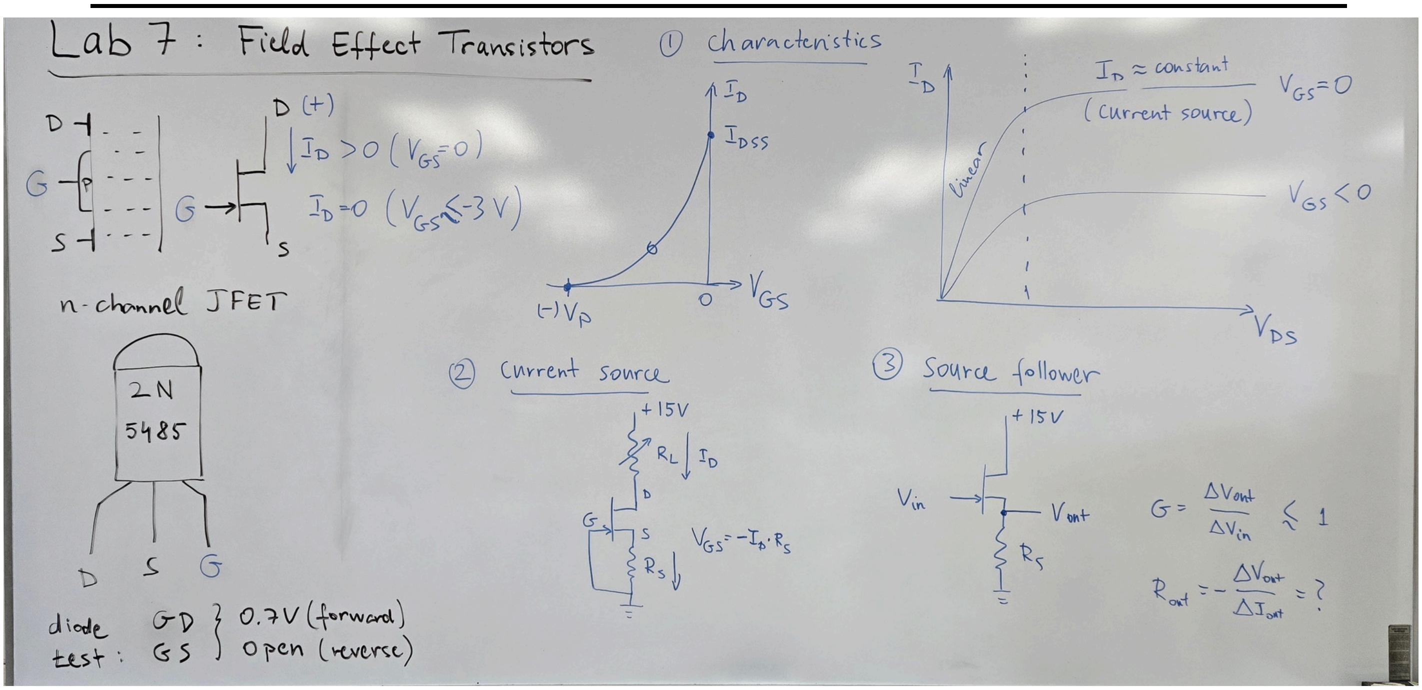
Equipment: breadboard, DMM's (2), function generator, oscilloscope, JFET (2N5485), decade resistor box, capacitors, resistors

Preliminaries:
As the name suggest, field-effect transistors work by controlling a current with an electric field produced by a voltage applied to one of its leads, rather than with a base current in the case of a bipolar transistor. A common type of FET is the junction field-effect transistor (JFET) which is available in two varieties: n-channel and p-channel. In an n-channel JFET, a conducting channel is created using n-doped semiconductor bar with two electric contacts, called drain and source, attached to its opposite ends. In the middle of the channel a p-doped semiconductor is diffused to form the gate. A voltage applied to the gate controls the number of charge carriers in the conducting channel.

The rule of thumb for normal operation of an n-channel JFET (polarities have to be reversed for p-channel JFET):

(1) The drain is held more positive than the source ($V_D > V_S$).
(2) The JFET is ON, i.e. current flows from the drain to source, when the gate voltage is the same as the source voltage ($V_G = V_S$).
(3) The drain-source current decreases as the gate voltage becomes more negative relative to the source voltage and the JFET eventually turns OFF, i.e. the drain-source current will stop, below the pinch-off voltage of about $V_{GS} \approx -3$V.

Note that the JFET is always operated with its gate reverse biased relative to the source (and drain), so there is no current flowing through the gate (except a tiny leakage current). This results in a super high input impedance (as large as $10^{14}\Omega$) at the gate of the transistor.



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Part 1: JFET Characteristics

The circuit in Figure 2 will be used to measure several characteristics of JFET transistors. Use two DMMs to measure the gate-to-source voltage V_{GS} (use Agilent 34450A DMM) and the drain current I_D (use the Keithley 179A DMM).

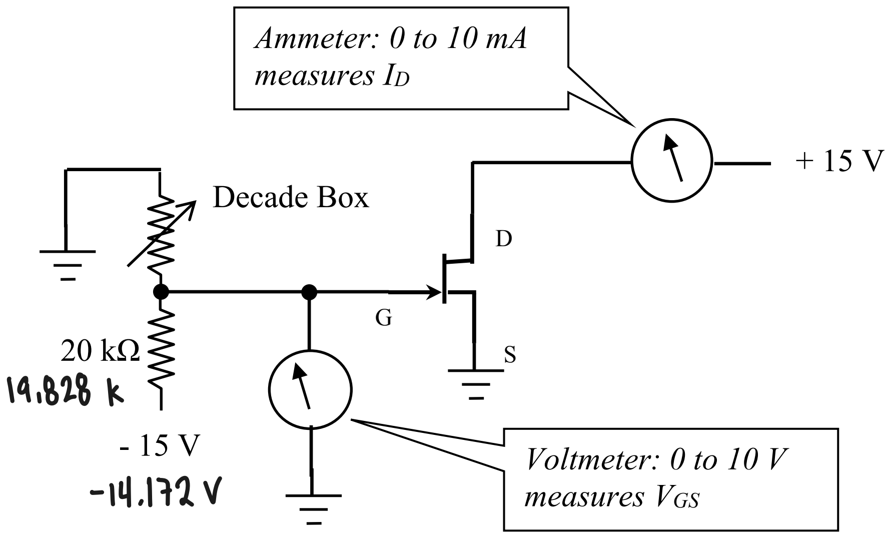

Figure 2

Note that the decade resistor box is part of a voltage divider that controls the gate voltage. The pinch-off voltage V_P is the (negative) gate-to-source voltage ($V_{GS} = V_G - V_S$) at which I_D becomes zero. Measure V_P by observing I_D while changing V_{GS}.

Pinch-off voltage V_P = (-)__-2.6274__ [V]

The maximum value of the current I_D when V_{GS} = 0 (gate and source connected) is usually called I_{DSS}. Measure and record its value.

I_{DSS} = __7.213__ [mA]

Measure the drain current I_D as a function of V_{GS} (from V_P to zero) and plot it.

V_{GS}[V]	0	0.245	0.616	1	1.238	1.529	1.807	2	2.23	2.427	2.627
I_D[mA]	7.213	6	4.32	2.79	1.96	1.085	0.443	.156	.023	.002	0

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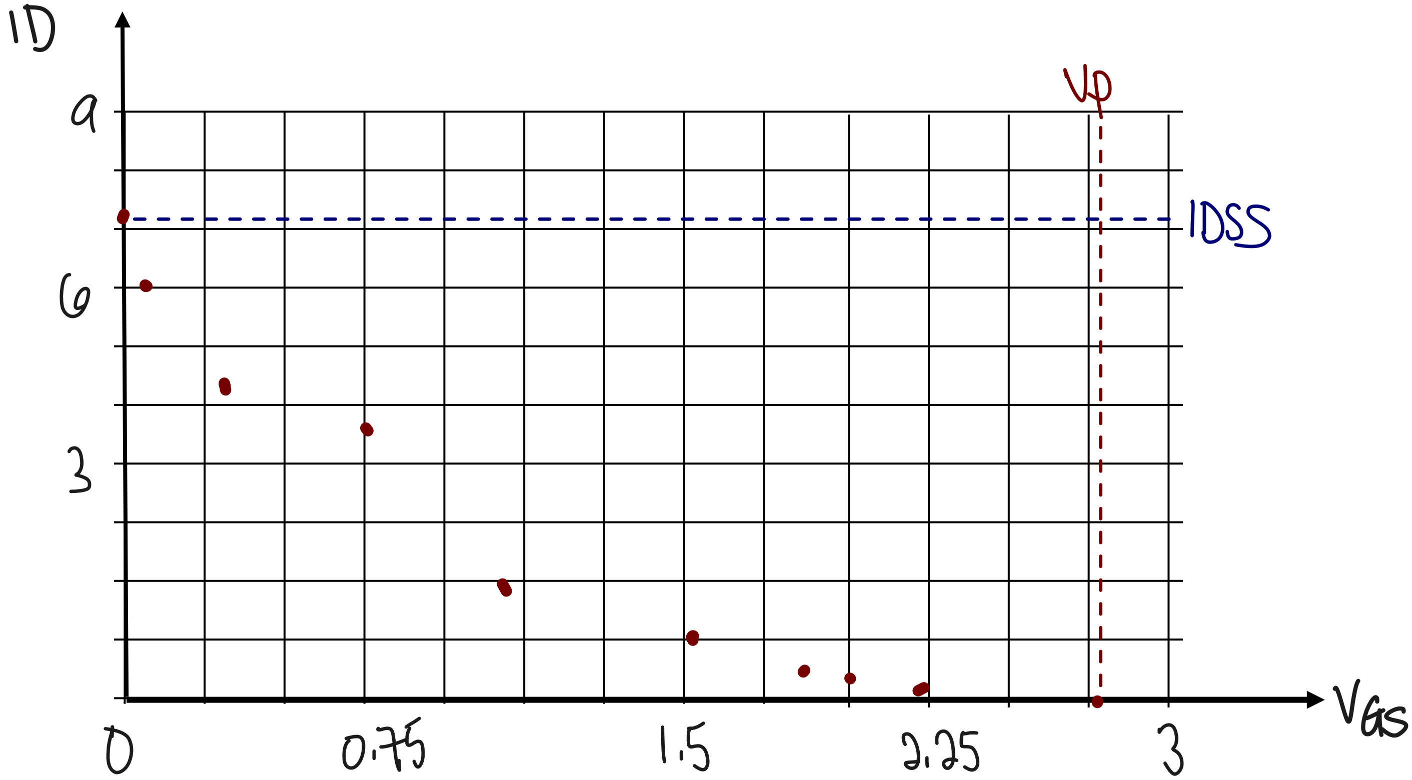
Plot I_D as a function of V_{GS}. Mark the special values V_p and I_{DSS} on the graph.



Part1b. Now measure the drain current (I_D) as a function of the drain-source voltage (V_{DS}) by changing the setup to that on Figure 3. Note that the gate-to-source voltage (V_{GS}) is now fixed to 0V. Use the second DMM to measure V_{DS} between the drain and the source of the transistor (not shown in the figure).

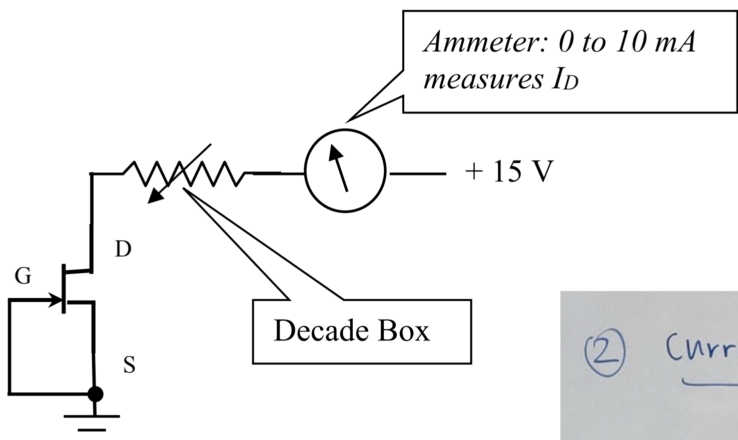
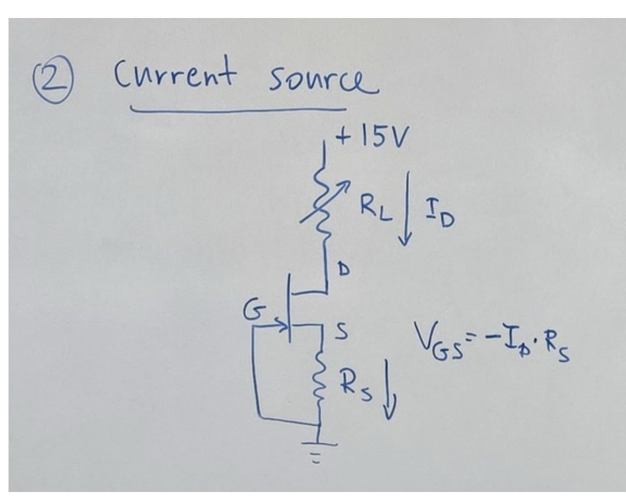


Figure 3



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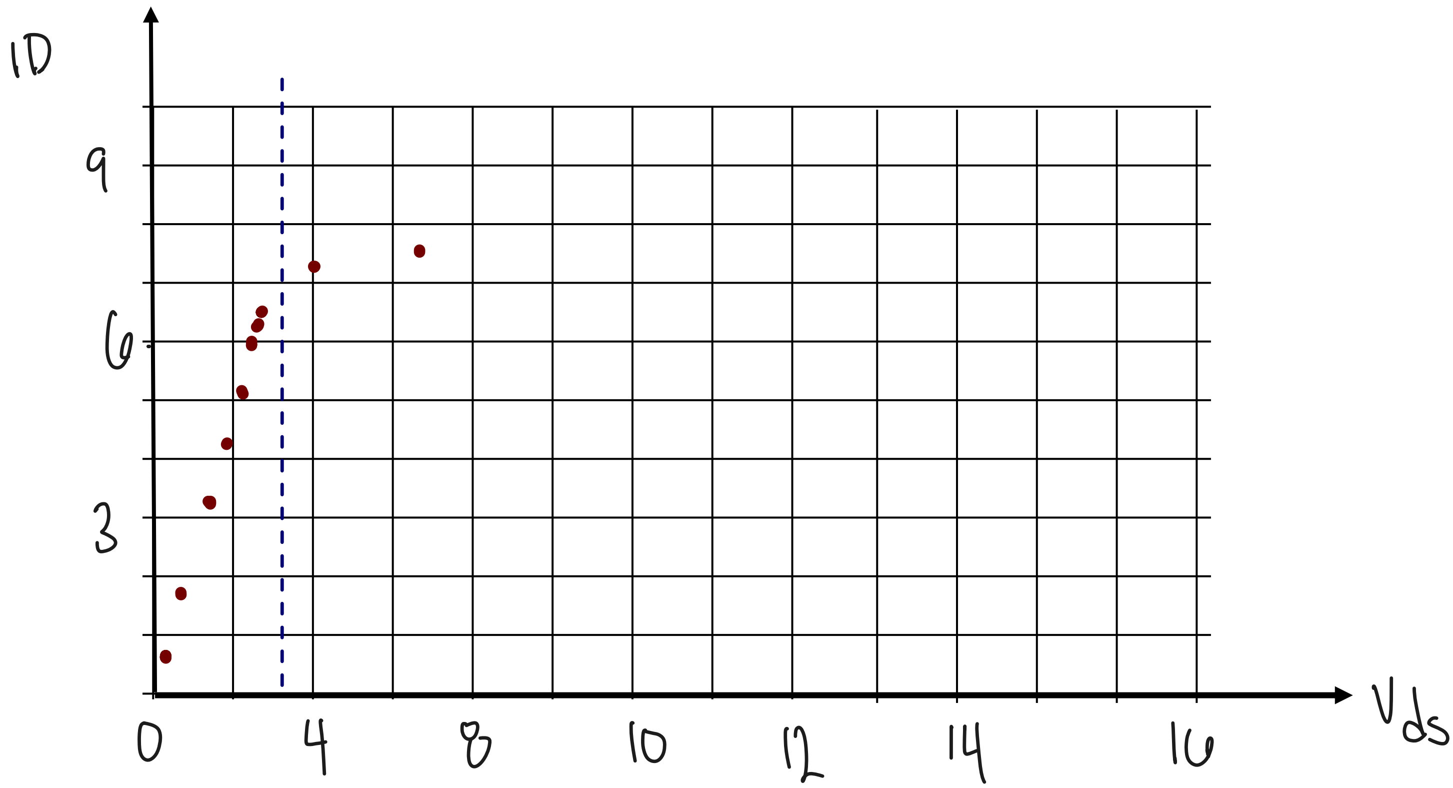
This time the variable resistor controls V_{DS} (and I_D). Record a few measurements with the resistor between 0 and 20 kΩ and plot the data. Make sure you make more frequent measurements in the region where I_D changes fast (when $V_{DS} \leq 2.5V$).

Ω	0Ω	1k	1.5k	1.9k	2k	2.16K	2.36K	2.42k	3.3K	4.6	8.6	20
V_{DS}[V]	14.83	7.28	4.2	2.41	2.12	1.79	1.50	1.19	0.91	0.60	0.30	0.12
I_D[mA]	7.54	7.34	7.13	6.54	6.35	6.05	5.64	5.026	4.23	3.10	1.69	0.74

Plot I_D as a function of V_{DS}:



You will notice that the current (I_D) through the transistor remains relatively constant for a range of resistor values, but then it starts to decrease fast as V_{DS} is squeezed (approaching 0V).

What is the approximate value of the resistor box when the current starts to drop fast?

1.5kΩ

Do not take the circuit apart! Keep it for Part 2.

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Part 2: FET constant current source

In the previous exercise you saw that the current through the transistor (I_D) remains almost independent of the "load" resistor (represented by the variable resistor connected to the drain) for a range of resistance values. By adding a resistor between source and gate to the previous circuit as shown in Figure 4, we can improve the performance of the current source and make the current independent of the load for a wider resistance range. Build the circuit and measure the drain current (I_D) as a function of the load resistor between 0 and 100 kΩ.

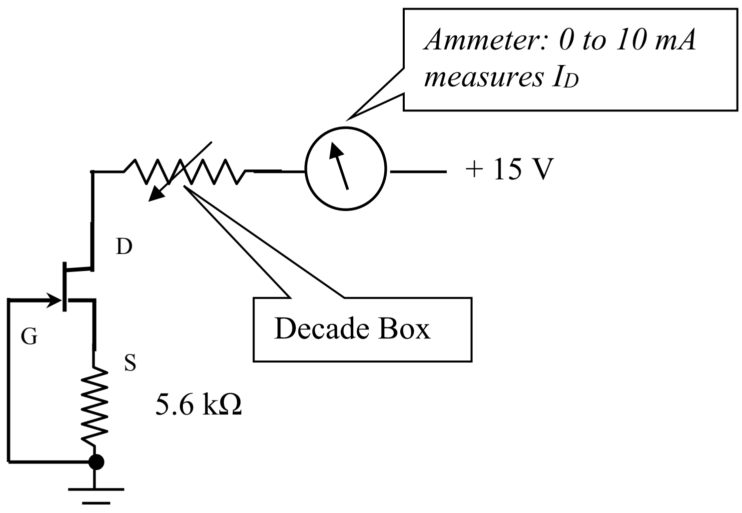
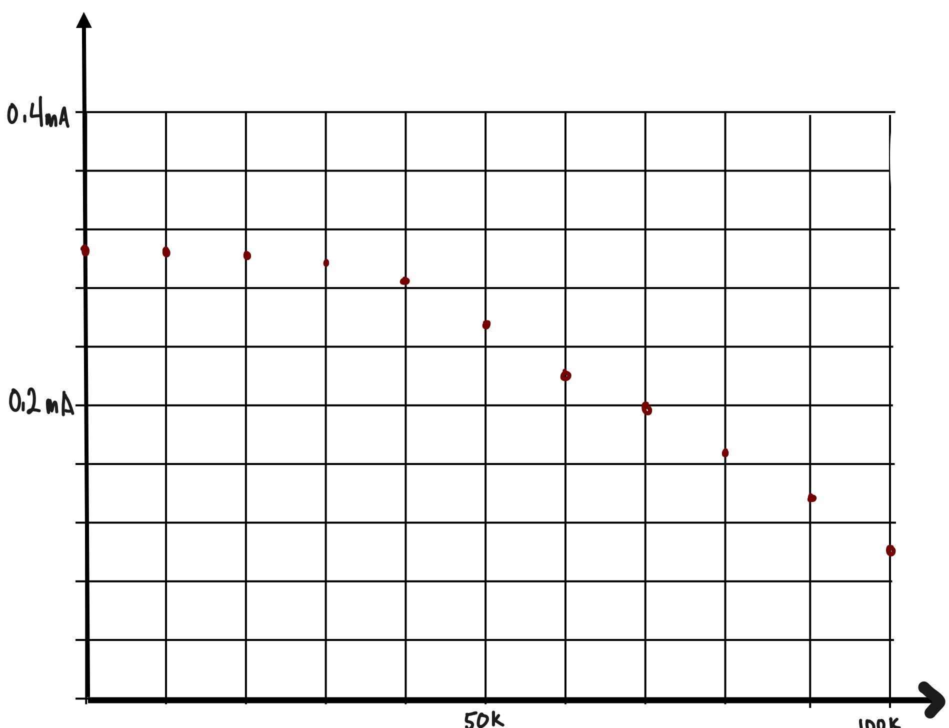


Figure 4



What is the resistance value where the current starts to decrease fast? How does it compare to the value at the end of Part 1b?

30K

much higher than 1b

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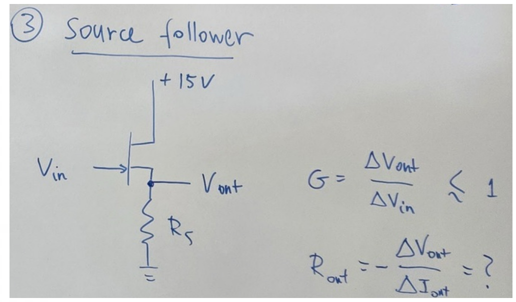
Part 3: FET source follower

Follower circuits of the kind shown in Figure 5 have a voltage gain (ratio of output/input voltage) of slightly less than unity. However, they are very useful because their output impedance is far lower than the input impedance.



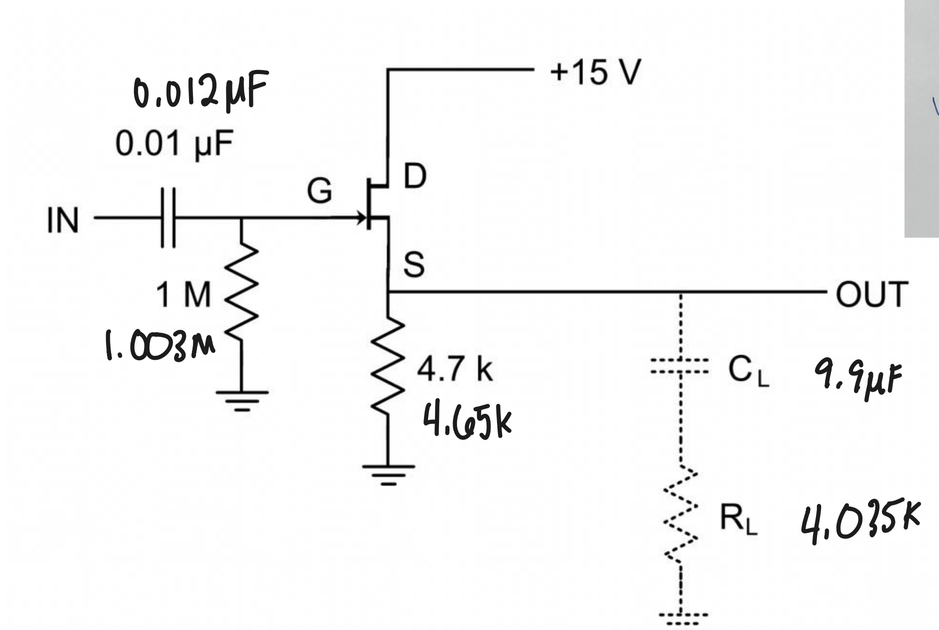


Figure 5

Build the circuit shown (without C_L and R_L) and test it with a small-amplitude sine wave of about 10 kHz. Display both input and output on the two-channel oscilloscope.

Measure the voltage gain of the follower (ratio of output and input amplitude): $G = \frac{\Delta V_{out}}{\Delta V_{in}}$

Voltage gain = 0.8

Increase the input amplitude until the output waveform starts to clip. Estimate the maximum, undistorted output amplitude.

Maximum undistorted output amplitude = 2.5V [V]

The input impedance of the circuit is essentially equal to 1 MΩ.

Use small amplitude input again and measure the output impedance by loading the circuit with a DC blocking capacitor C_L approximately 10 μF (positive terminal connected to the FET source!) and resistive load R_L on the order of 4.7 kΩ or lower. The reduction of the output amplitude when adding the load allows you to calculate the output impedance.

Output amplitude without load: .876 [V]

Output amplitude with load: .776 [V]

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Calculate the output impedance of the circuit.

$$V_L = V_{no_L} \cdot R_L / (R_L + R_{out})$$

$$R_{out} = R_L\left(V_{no_L} / (V_L - 1)\right)$$

$$= 4.035\,k\Omega \cdot (0.876/0.476 - 1)$$

$$= 3.391\,k\Omega$$

Output impedance: ____3391____ [Ω]

Done